Scalable Technologies for Data Analysis

Introduction / Presentation

Davide Carneiro

Mestrado em Engenharia Informática

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Scalable Technologies for Data Analysis

"*tekhne*" (technique, arts, craft) + "*logia*" (study of)
A product of science and engineering that involves a set of instruments, methods and techniques aimed at solving problems.

Scalable Technologies for Data Analysis

1. That can be climbed (e.g., climbable wall).
2. That is able to grow uniformly or to withstand an increase in load.

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Scalable Technologies for Data Analysis

A process through which data is inspected, cleaned, transformed, and modeled in order to discover useful information, suggest conclusions, and support decision-making processes

1. The big (data) picture

Literally...

THE 2025 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE

MAD.FIRSTMARK.COM

APPLICATIONS & AGENTS

ENTERPRISE

- CUSTOMER SERVICE & SUPPORT AGENTS
- SALES & MARKETING AI (GTM)
- AUTOMATION & OPERATIONS (AGENTIC RPA)
- HUMAN CAPITAL & TALENT
- LEGAL AI
- FINOPS & RISK

HORIZONTAL

- CODE AI & AGENTS
- TEXT, DOCS & KNOWLEDGE AGENTS
- MEDIA GENERATION
- 3D / ANIMATION / GAMING
- AUDIO & VOICE AGENTS

INDUSTRY

- HEALTHCARE & LIFE SCIENCES
- FINANCE
- ROBOTICS, AUTONOMY & INDUSTRIAL
- AEROSPACE, DEFENSE & GOVTECH

INFRASTRUCTURE

STORE & STREAM

- STORAGE
- DATA LAKES/LAKEHOUSES
- DATA WAREHOUSES
- EVENT BROKERS & MESSAGING
- STREAM PROCESSING & STREAMING ETL
- RELATIONAL & DISTRIBUTED SQL
- NOSQL DATABASES
- REAL-TIME ANALYTICS / HTAP
- GRAPH DATABASES
- VECTOR DATABASES
- GPU DATABASES
- MULTI-MODEL DATABASES & ABSTRACTIONS

INTEGRATE & OPERATE

- ETL / ELT / DATA TRANSFORMATION
- UNSTRUCTURED & VECTOR ETL
- DATA INTEGRATION
- DATA GOVERNANCE & CATALOG
- ORCHESTRATION
- DATA QUALITY & OBSERVABILITY
- CLOUD OBSERVABILITY
- PRIVACY & SECURITY
- COMPUTE

ANALYTICS

- BI & AI ANALYTICS PLATFORMS
- HEADLESS / EMBEDDED BI & SEMANTIC LAYER
- EXPERIMENTATION & CAUSAL
- PRODUCT ANALYTICS

MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

BUILD

- AI DEVELOPER PLATFORMS
- LOCAL/ON-DEVICE LLM RUNTIMES
- AGENT PLATFORMS
- AGENT INFRA / TOOLING

VALIDATE & SECURE

- AI OBSERVABILITY & EVALUATION
- AI SAFETY & SECURITY
- ROUTING, PROMPT MGT & EXPERIMENTATION
- AI & AGENT GOVERNANCE

OPERATE & SUPPLY

- ML OPS
- DATA GENERATION & LABELING
- DATA SCIENCE
- ENTERPRISE AI PLATFORMS

MODALITIES & RESEARCH

- SPEECH & VOICE AI
- COMPUTER VISION
- AI R&D — COMMERCIAL
- AI R&D — NONPROFIT / ACADEMIC

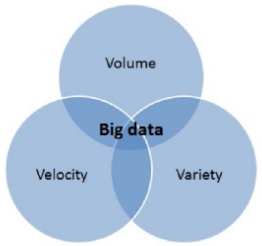
RUN & DEPLOY

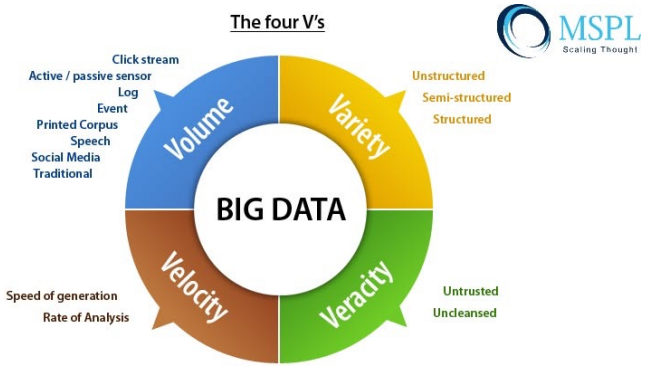
- GPU CLOUD / DEPLOYMENT INFRA
- MODEL SERVING & INFERENCE RUNTIMES
- AI HARDWARE
- EDGE AI

DATA SOURCES & APIS

- DATA MARKETPLACES & DISCOVERY
- WEB SEARCH & CRAWL APIS
- FINANCIAL & ALTERNATIVE DATA
- GEOSPATIAL, SATELLITE & LOCATION
- COMPANY / PEOPLE / ENTITY ENRICHMENT

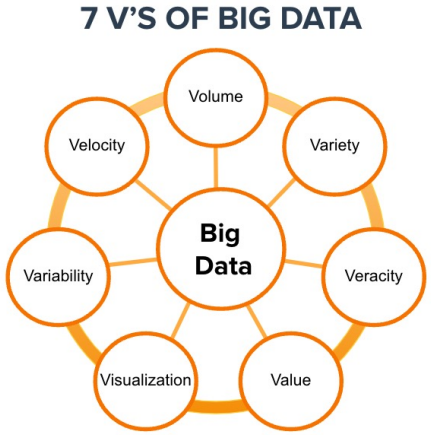
DATA & AI CONSULTING


Volume
Big data
Velocity
Variety

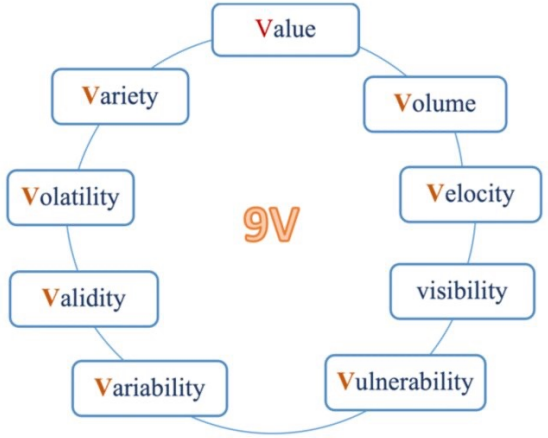

The four V's
MSPL
Scaling Thought
Click stream
Active / passive sensor
Log
Event
Printed Corpus
Speech
Social Media
Traditional
Volume
Variety
BIG DATA
Velocity
Veracity
Unstructured
Semi-structured
Structured
Speed of generation
Rate of Analysis
Untrusted
Uncleansed

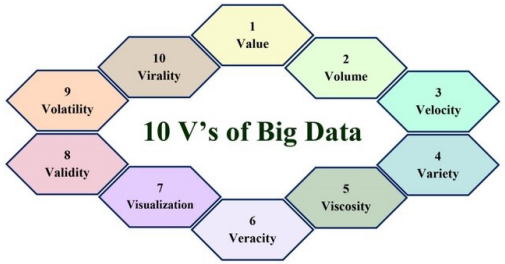

Volume
Value
Big Data
Velocity
Veracity
Variety

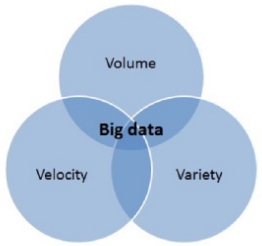

Value
Volume
Veracity
Big Data
Velocity
Variety
Variability

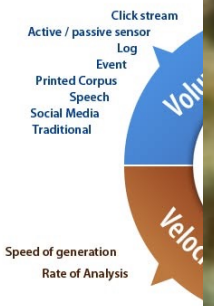

7 V'S OF BIG DATA
Volume
Velocity
Variety
Big Data
Variability
Veracity
Visualization
Value

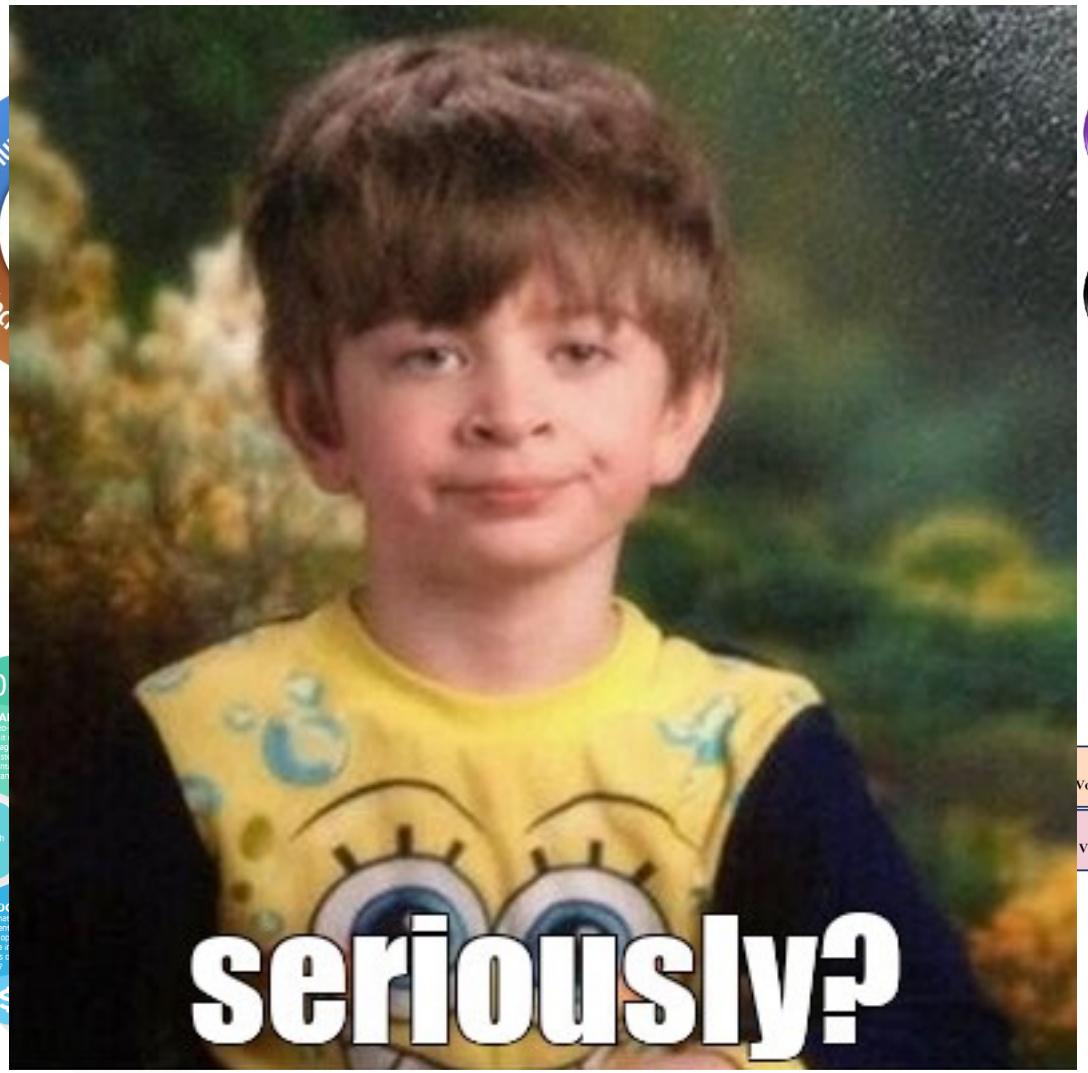

01
VOLUME
02
VALUE
03
VERACITY
04
VISUALISATION
05
VARIETY
06
VELOCITY
07
VISCOSITY
08
VIRALITY
BIG DATA
with 8 V's


Value
Variety
Volume
Volatility
9V
Velocity
Validity
visibility
Variability
Vulnerability

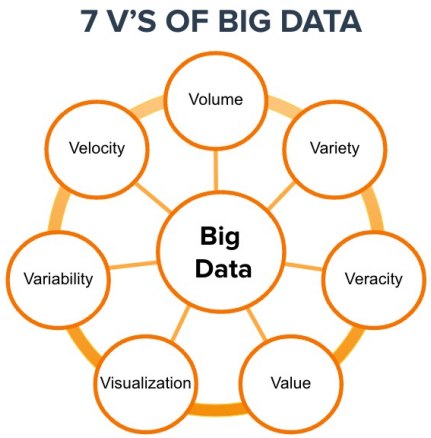

1 Value
10 Virality
2 Volume
9 Volatility
3 Velocity
10 V's of Big Data
8 Validity
4 Variety
7 Visualization
5 Viscosity
6 Veracity


Volume
Big data
Velocity
Variety


Click stream
Active / passive sensor
Log
Event
Printed Corpus
Speech
Social Media
Traditional
Speed of generation
Rate of Analysis


seriously?


Value
Volume
Veracity
Big Data
Velocity
Variety
Variability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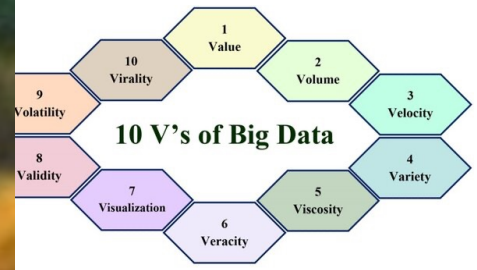
Visualization
Value

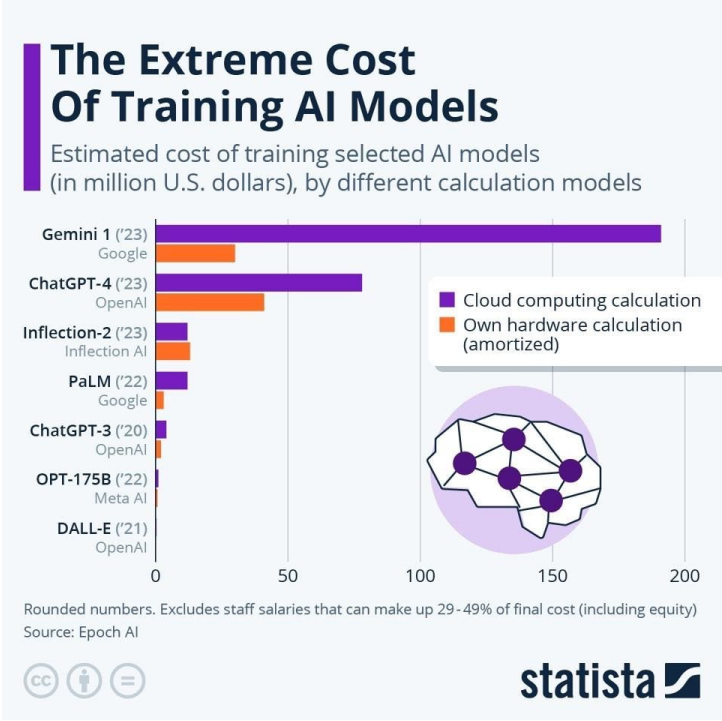

07
VISCOSITY
06


10 V's of Big Data
1 Value
2 Volume
3 Velocity
4 Variety
5 Viscosity
6 Veracity
7 Visualization
8 Validity
9 Volatility
10 Virality

The core problem

- Scalability is not just “more data”
  - **Volume**: data grows faster than teams
  - **Velocity**: updates are continuous, not monthly
  - **Variety**: logs, events, tables, files, APIs

- There are now real-world constraints and requirements
  - Cost
  - Reliability
  - Governance
  - Security
  - …

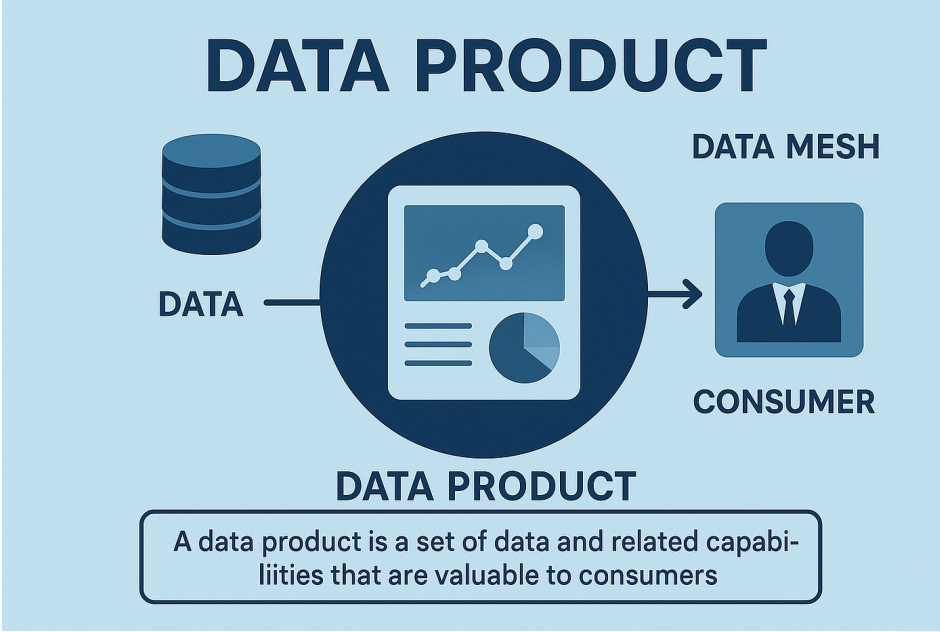


What does scalable mean? Why companies care?

- A scalable solution
  - Keeps working as data + users grow
  - Has predictable cost/performance trade-offs
  - Is observable + debuggable in production
  - Makes changes safely (schemas, pipelines, metrics)
- Scalability is technical + organizational
- Scalable data platforms enable
  - Faster decision cycles from operations to strategy
  - Consistent KPIs, i.e."one version of truth"
  - Reliable data products (e.g. recommendations, risk, forecasting)
  - Regulatory + audit requirements (who accessed what, when)

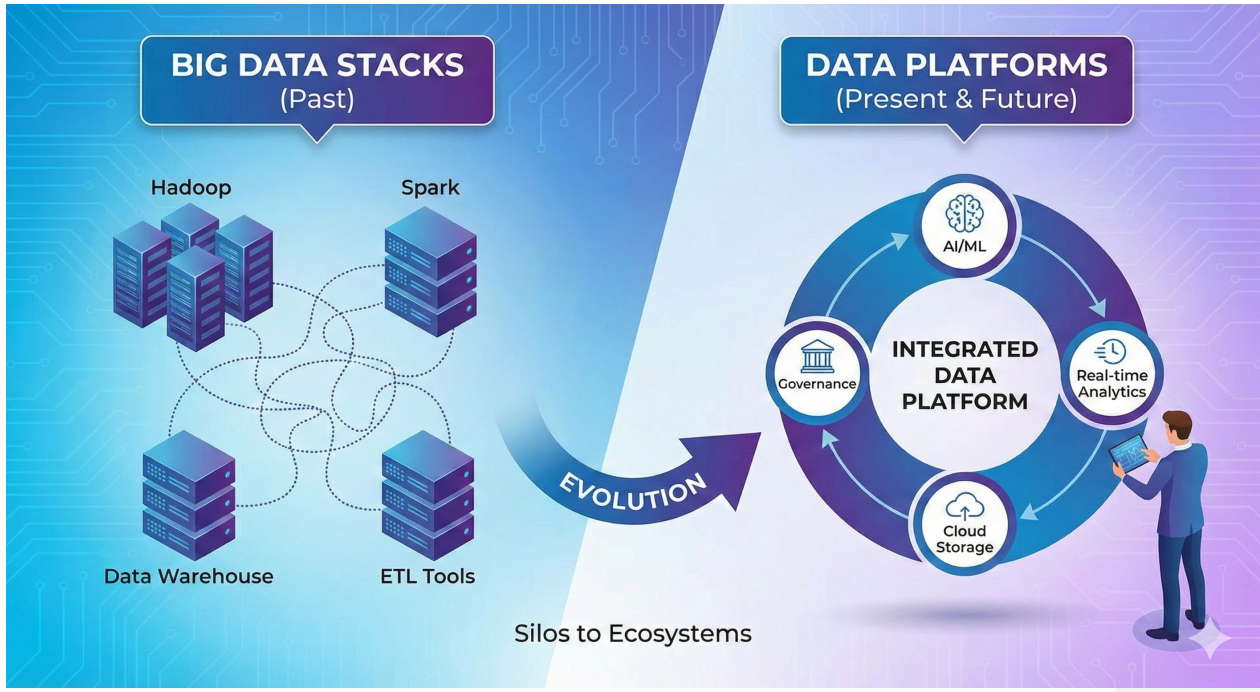
From outputs to data products

- We'll treat data as a product, not as a generic dataset
  - Defined consumers + use cases
  - Clear semantics (metrics, dimensions)
  - Contracts and SLAs/SLOs
  - Versioning, documentation, ownership



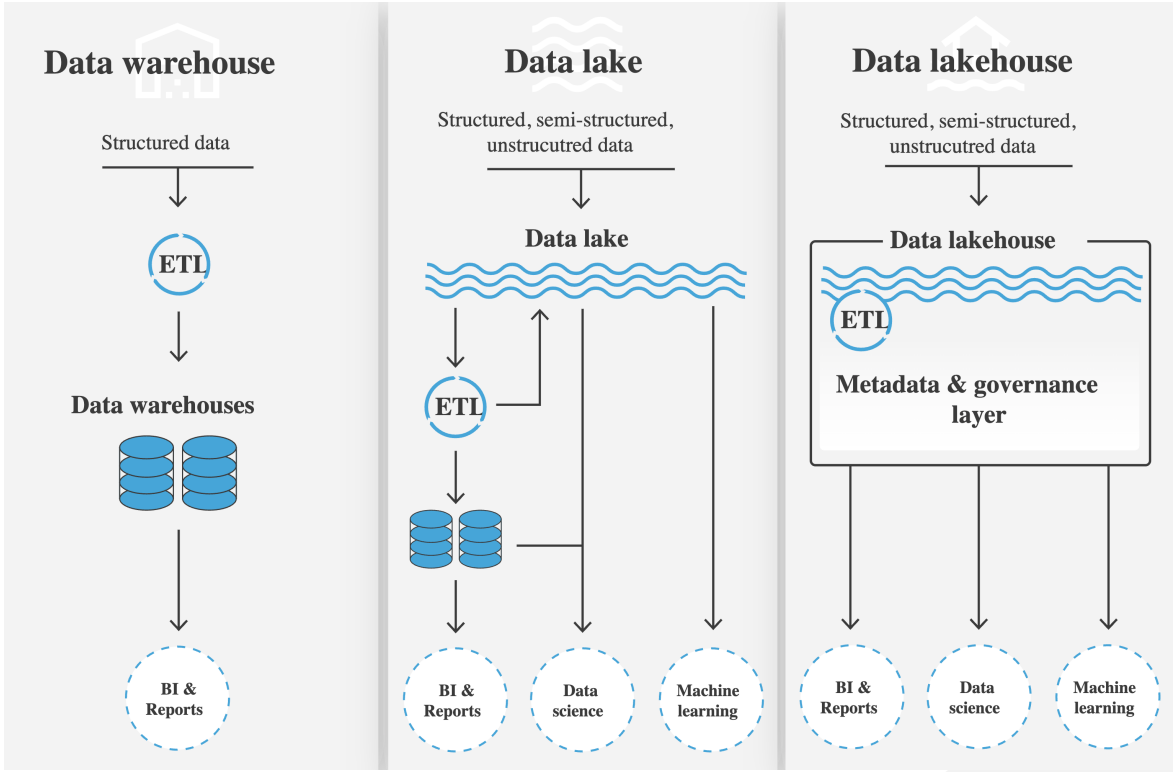
From Big Data stacks to data platforms

- First: ad-hoc scripts + single DB
- Then: “Big Data” ecosystems
- Now: data platforms / modern data stacks
  - Separation of compute/storage
  - Modular engines (batch, SQL, streaming)
  - Governance + observability built-in
- TEAD is also changing...



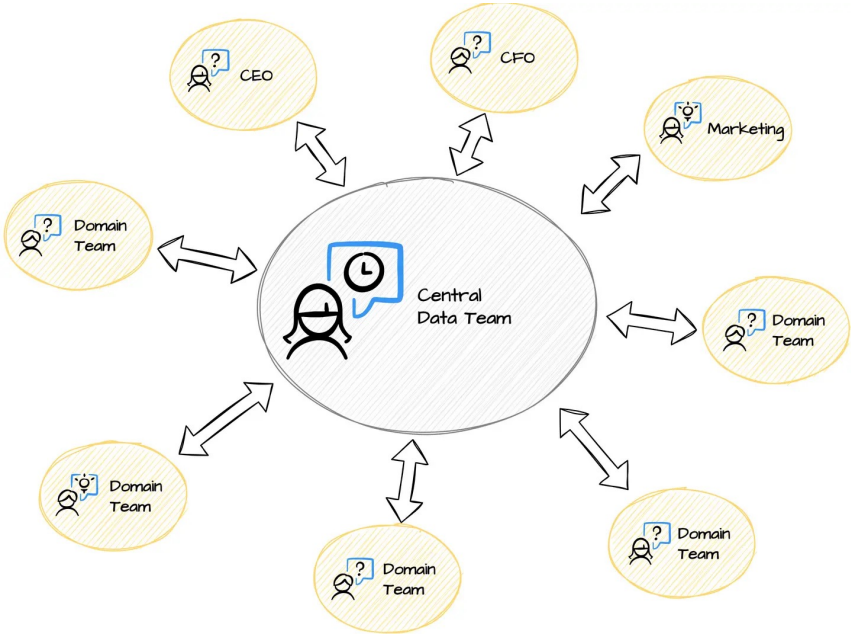
Reference architectures

- **Data Warehouse**: A highly structured repository optimized for fast SQL queries and business intelligence using cleaned, pre-processed data
- **Data Lake**: A low-cost, flexible storage space that holds massive amounts of raw data in its native format for data science and large-scale processing
- **Data Lakehouse**: A modern hybrid that applies the performance and governance of a warehouse directly onto the scalable, low-cost storage of a lake

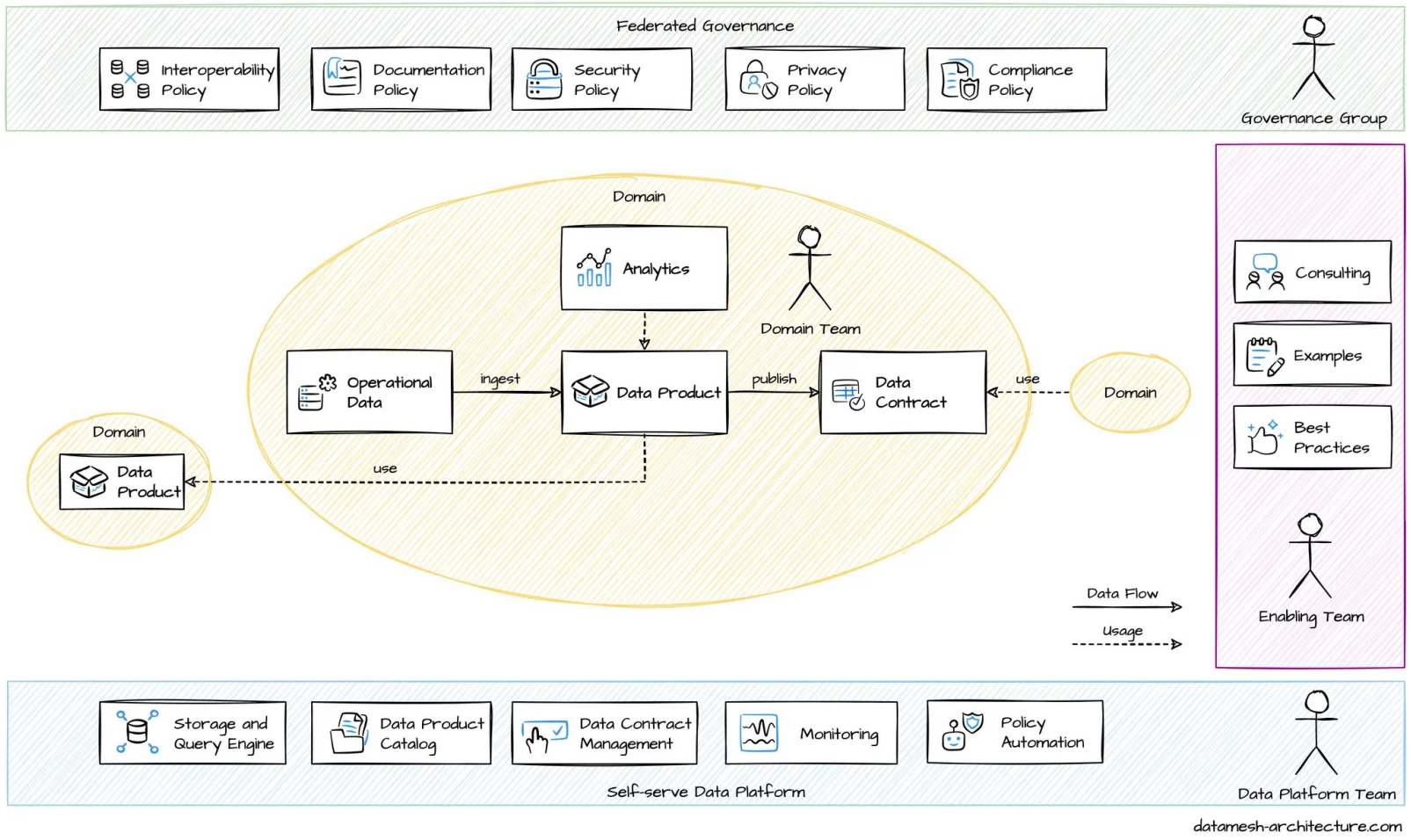


More recently… Data Mesh

- While a Data Lakehouse is a technology (how to store and query data), Data Mesh is a strategy (how to manage and own data across a large company)
  - Domain-Oriented Ownership: Instead of being centralized, each business unit (e.g., Sales, Marketing, HR) owns and manages its own data (although with a central infrastructure)
  - Data as a Product: Data is treated with the same care as a customer-facing product, meaning it must be discoverable, reliable, and easy to use for others in the company
  - Self-Serve Data Infrastructure: A central platform team provides the tools (like a Lakehouse) so that business units can manage their data without needing to be IT experts
  - Federated Computational Governance: Global standards (like security and naming conventions) are automated across all domains to ensure everything stays compatible.



Data Mesh Architecture



https://www.datamesh-architecture.com/

2. About us

Learning outcomes

- By the end of this course, you should be able to:
  - Identify when scalability + non-functional requirements change the architecture
  - Choose between lake/warehouse/lakehouse based on trade-offs
  - Design pipelines with contracts, versioning, and layered data models
  - Optimize processing (shuffle, skew, partitions, materialization choices)
  - Operate production-grade pipelines (idempotency, retries, backfills, observability)
  - Integrate data quality, governance, and (optionally) ML components

What you should be able to avoid

- After finishing this course you should be able to prevent common causes of “data pain”
  - Pipelines that break silently
  - Dashboards that disagree
  - “Hero engineers” + tribal knowledge
  - Costs that explode with no explanation
  - Changes that break downstream consumers

Industry roles you're training for

- Data Engineering
  - Focuses on building and maintaining the foundational infrastructure and automated pipelines that move raw data reliably and at scale from sources to storage
- Analytics Engineering
  - Operates at the intersection of data and business, transforming raw data into clean, documented, and reusable models within a semantic layer for consistent reporting
- Data Platform
  - Provides the centralized toolkit and ecosystem (including orchestration, governance, and cost monitoring) that enables data and analytics engineers to work efficiently and securely

How the course works

- Theory/practice sessions: concepts + patterns + architectural decision-making
  - Concepts + reference architectures
  - Case discussions (trade-offs: cost/latency/reliability/security)
  - Guided problem-solving micro-exercises
  - Short demos (illustrating concepts, not tool-centric)
- Labs: hands-on, project-based implementation
  - A longitudinal scenario
  - Implement an end-to-end scalable architecture (containerized)
  - Orchestrate production-grade pipelines
  - Validate with evidence: performance, quality, reliability, reproducibility
- Evaluation
  - 60% group project (9.5+)
  - 40% written test (at the same date of the exam) (7.5+)

The plan (high-level view)

1. Contract-Driven Design and Data Flow Modeling
2. Storage + table formats
3. Processing patterns + optimization
4. Orchestration + production reliability
5. Serving: SQL/APIs/semantic layer
6. Quality + observability + cost
7. Incremental + streaming
8. ML integration
9. Governance + evolution

1. Contract-Driven Design and Data Flow Modeling

- From the problem to analytical questions and metrics (data products)
- DAGs, determinism, and reproducibility in data pipelines
- Data contracts: schemas, keys, semantics, granularity, SLAs/SLOs
- Schema evolution and versioning; change management and compatibility
- Bronze/silver/gold layers and responsibilities per layer

2. Storage + table formats

- Object storage vs. filesystem vs. databases: operational and cost trade-offs
- Columnar formats (Parquet/ORC), compression, and pushdown
- Partitioning, clustering, and file management (small files, compaction)
- Tables and metadata in a lakehouse (Iceberg/Delta/Hudi, concepts and use cases)
- Catalog, metadata, and lineage (concepts and practical implications)

3. Processing patterns + optimization

- Transformation patterns: filters, aggregations, joins, windowing
- Shuffle, skew, parallelism, and mitigation strategies
- Materialization vs. query-on-read, caching and persistence management
- Measurement and evidence: profiling, execution plans, and job metrics
- Prototyping vs. production: limits, risks, and best practices

4. Orchestration + production reliability

- Orchestration vs. processing: responsibilities and limits
- Flyte concepts: tasks, workflows, launch plans, packaging, and execution
- Idempotency, retries, timeouts, backfills, and temporal parameterization
- Environment configuration, secrets, and reproducibility
- Versioning, caching, and rapid iteration, execution traceability

5. Serving: SQL/APIs/semantic layer

- Serving patterns: marts, OLAP vs. OLTP, distributed query engines
- SQL as an interface: metric governance and reuse
- Serving via query engines (e.g. Trino) vs. database materialization (e.g. Postgres)
- APIs and data products: contracts, versioning, and limits
- Visualization and reporting: design principles for analytical consumption

6. Quality + observability + cost

- Data quality: dimensions, per-layer checks, validation gates
- Pipeline testing: unit/integration/contract tests, synthetic data
- Observability: structured logs, metrics, tracing, and runbooks
- Incident management: data downtime, severity, and response
- Performance and cost: evidence-driven tuning, cost per pipeline/query

7. Incremental + streaming

- Incrementality: snapshots, upserts/merge, deduplication, logical watermarking
- CDC (Change Data Capture) and pipeline integration, handling late data
- Streaming: concepts (event time vs. processing time), delivery semantics
- Event-driven architectures (e.g. Kafka/Redpanda – overview)
- Criteria for choosing batch vs. incremental vs. streaming

8. ML integration

- Feature engineering and temporal validation, leakage and baselines
- Training reproducibility and pragmatic evaluation
- Batch scoring vs. online serving: trade-offs
- Tracking and artifact management (Minimum Viable MLOps)
- Integrating ML into the orchestrated pipeline (e.g. MLFlow)

9. Governance + evolution

- PII, compliance, and retention policies, auditing
- Access control and environment segregation
- Lineage and documentation as a data product component
- Change management: migrations, breaking changes, and communication
- Roadmap, technical debt, and platform maturity criteria

The stack

	Technology	Role	Key Syllabus Modules
	Minio	The physical "bucket" where all raw, transformed, and model data lives.	Scalability, Object Storage
	Apache Iceberg	Adds a database-like layer to S3. Handles schema evolution, ACID transactions, and time travel	Contracts, Lakehouse, Incrementality
	Trino	The high-speed SQL layer. It allows Analytics Engineers to query Iceberg tables across S3 without moving data	Query-on-read, Serving, Cost/ Performance
	Flyte	Manages the DAGs. It ensures pipelines are reproducible, idempotent, and can handle complex ML workflows	DAGs, Orchestration, ML Integration
	MLflow	Tracks experiments, versions models, and manages the model registry	Feature Engineering, Tracking, MLOps

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